

4th Semester		
ISCED Code	Course Code	Course Title
0613	CSE-2421	Computer Algorithms
Credit Hours: 3	Contact Hours: 3	Type: Core, Engineering
Prerequisite: CSE-2321 (Data Structures)		
Co-requisite: CSE-2422 (Computer Algorithms Lab)		

Course Assessments	CIE: Continuous Internal Evaluation	Attendance	10 Marks
		Class test/ Assignment/ Quizzes	10 Marks
		Mid-term	30 Marks
	SEE: Semester End Examination		50 Marks
	Total		100 Marks

Course Rationale:

This course introduces students to the general tools and techniques for analyzing and designing computer algorithms. Initially necessary mathematical preliminaries required for analyzing and designing computer algorithms are taught. Then this course familiarizes students with several algorithmic approaches and corresponding problems. To take this course, students should be able to program in a standard programming language preferably in C/C++. They should have familiarity with basic data structures. Some mathematical maturity also is expected; students should have some idea of what constitutes a mathematical proof and how to write one. Some knowledge of basic probability will also be helpful. Knowledge of this course will be needed as prerequisite knowledge for future courses such as Artificial Intelligence, Computer Networks and Compiler.

Objectives:

Upon completion of the course, students will be able to:

1. Demonstrate a familiarity with major algorithms and data structures.
2. Analyze algorithms for understanding and comparing their working mechanism.
3. Analyze algorithms for deriving proof of correctness.
4. Analyze the asymptotic performance of algorithms.
5. Analyze and apply important algorithmic design techniques and methods of analysis.

Course Learning Outcomes (CLOs):

Upon successful completion of this course, students will be able to:

#	CLO Description	Weightage (%)
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1.	Demonstrate a familiarity with major data structures and sorting, searching and tree-based algorithms and algorithms used in different application areas.	30 %
2.	Analyze the time and space complexity of algorithms, demonstrate familiarity with different complexity classes and their implication and construct rigorous correctness proofs for algorithms using formal proof techniques.	10 %
3.	Apply graphs to model engineering problems, when appropriate and demonstrate a familiarity with major graph algorithms and their analyses.	30 %
4.	Design and implement efficient algorithms using algorithmic paradigms such as divide and conquer, greedy methods, dynamic programming, and backtracking and demonstrate a familiarity with major algorithms that employ these paradigms.	30 %

Mapping of CLO-PLO-DL-KP-EP-EA:

#	CLOs	PLOs	DL	KP	EP	EA	Teaching Learning Strategy (TLS)	Assessment Strategy (AS)
CLO1	Demonstrate a familiarity with major data structures and sorting, searching and tree-based algorithms and algorithms used in different application areas.	PLO1	C2	K3	-	-	Lectures Problem Solving	Class Text Mid Exam Final Exam Quiz
CLO2	Analyze the time and space complexity of algorithms, demonstrate familiarity with different complexity classes and their implication and construct rigorous correctness proofs for algorithms using formal proof techniques.	PLO2	C5	K2 , K3	-	-	Lectures Problem Solving	Class Text Mid Exam Final Exam Quiz
CLO3	Apply graphs to model engineering problems,	PLO2	C4	K3	-	-	Lectures	Class Text Mid Exam

	when appropriate and demonstrate a familiarity with major graph algorithms and their analyses.						Problem Solving	Final Exam Quiz
CLO4	Design and implement efficient algorithms using algorithmic paradigms such as divide and conquer, greedy methods, dynamic programming, and backtracking and demonstrate a familiarity with major algorithms that employ these paradigms.	PLO2	C4	K3	-	-	Lectures Problem Solving	Class Text Mid Exam Final Exam Quiz
Note: DL: Domain/level of learning taxonomy, KP: Knowledge Profile, EP: Attribute of Complex Engineering Problems, EA: Attribute of Complex Engineering Activities, Learning Domains (C: Cognitive, A: Affective, P: Psychomotor)								

Course Content:

#	Content	Duration	CLOs
Mid-Term (30 Marks)			
1	Algorithm and Data structure: Introduction to algorithm; Properties of good algorithm; Correctness proof and techniques for time and space complexity analysis of algorithms (using insertion-sort as an example); Application areas of algorithm; Growth of functions and asymptotic notations	2 weeks	CLO1 – CLO4
2	Divide and Conquer approach Sorting and Heaps: Divide and Conquer approach; Asymptotic bound of recursive algorithm by solving recurrence, substitution method, recursion tree method, master method; Maximum subarray problem; Merge-sort and complexity analysis; Quick-sort, Randomized Quick-sort and complexity analysis; Heap, Heap construction algorithm, Heap-sort, Priority queue, complexity analysis of related algorithms; Sorting in Linear Time, Lower bound of sorting, Counting sort, Radix sort.	2 weeks	CLO1 - CLO4

3	Dynamic Programming: Dynamic programming, Elements of dynamic programming, Memoization, Matrix-chain multiplication and longest common subsequence problems as examples, Complexity analysis of the algorithms.	2 weeks	CLO1 - CLO4
Final Exam: 50 Marks Group-A (20 Marks)			
4	Greedy Algorithms and String Matching Algorithms: Greedy algorithms, Activity selection problem, Elements of greedy strategy, Huffman codes and its application; String Matching Algorithms, Naive string-matching algorithm, Rabin-Karp algorithm; Complexity analysis of the algorithms.	2 weeks	CLO1 - CLO4
5	Graphs Basic & Traversal Techniques: Representation of Graphs, Breadth First Search, Depth First Search, Algorithm of BFS and DFS, Application of BFS and DFS, Minimum Spanning Tree, Kruskal's and Prim's Algorithm, Complexity analysis of the algorithms.	2 weeks	CLO1 - CLO4
Group-B (30 Marks)			
6	Shortest Path Algorithms: Single-source shortest path, Dijkstra's Algorithm, Bellman-Ford's Algorithm; All-pairs shortest path, Floyd-Warshall's Algorithm; Complexity analysis of the algorithms.	2 weeks	CLO1 - CLO4
7	Computational Geometry & Number Theory: Computational Geometry, Line Segment Properties, Convex Hull, Graham Scan Algorithm of Convex Hull, Number Theory, GCD, Modular Arithmetic, Prime Number generation, Complexity analysis of the algorithms.	2 weeks	CLO1 - CLO4
8	Theory of NP-Completeness and Coping with Hardness: Theory of NP-Completeness, P, NP, NP-Complete and NP-Hard Problems; Backtracking, N-Queen Problem; Branch and Bound; Approximation algorithms.	1 weeks	CLO2

Weekly Activity Plan:

Week	Topic	TLS	AS	CLOs
Week 1	Introduction to algorithm; Properties of good algorithm; Correctness proof and techniques for time and space complexity analysis of algorithms (using insertion-sort as an example); Application areas of algorithm.	Lecture		

Week	Topic	TLS	AS	CLOs
Week 2	Growth of functions and asymptotic notations; Divide and Conquer approach; Merge-sort and complexity analysis.	Lecture		
Week 3	Asymptotic bound of recursive algorithm by solving recurrence, substitution method, recursion tree method, master method; Review.	Lecture	Quiz	
Week 4	Quick-sort, Randomized Quick-sort and complexity analysis; Heap, Heap construction algorithm, Heap-sort, Priority queue, complexity analysis of related algorithms.	Lecture		
Week 5	Sorting in Linear Time, Lower bound of sorting, Counting sort, Radix sort; Dynamic programming, Matrix-chain multiplication.	Lecture		
Week 6	Matrix-chain multiplication and longest common subsequence, Elements of dynamic programming, Memoization; Review	Lecture	Quiz	
Week 7	Greedy algorithms, Activity selection problem, Elements of greedy strategy, Huffman codes and its application.	Lecture		
Week 8	String Matching Algorithms, Naive string-matching algorithm, Rabin-Karp algorithm.	Lecture		
Week 9	Representation of Graphs, Breadth First Search, Depth First Search, Algorithm of BFS and DFS, Application of BFS and DFS.	Lecture		
Week 10	Minimum Spanning Tree, Kruskal's and Prim's Algorithm; Review	Lecture		
Week 11	Single-source shortest path, Dijkstra's Algorithm, Bellman-Ford's Algorithm.	Lecture	Quiz	
Week 12	All-pairs shortest path, Floyd-Warshall's Algorithm; Complexity analysis of the algorithms; Review	Lecture		
Week 13	Computational Geometry, Line Segment Properties, Convex Hull, Graham Scan Algorithm of Convex Hull	Lecture		
Week 14	Number Theory, GCD, Modular Arithmetic, Prime Number generation, RSA Review	Lecture	Quiz	
Week 15	Theory of NP-Completeness, P, NP, NP-Complete and NP-Hard Problems;	Lecture	Quiz	

Week	Topic	TLS	AS	CLOs
	Backtracking, N-Queen Problem; Branch and Bound; Approximation algorithms			

Marks Distribution:

Course Assessment Pattern (Theory courses):

		CIE (50 marks)			SEE (50 marks)
Cognitive learning	Affective learning	Attendance Marks (10)	Assignment/ Class Test (10)	Mid-term (30)	Written Exam (50)
Remember					
Understand				5	5
Apply			5	15	15
Analyze			5	10	15
Evaluation					15
Create					
	Responding				
Total allocated marks					

Grading Policy: As per IIUC grading policy

Learning Materials:

Text Books:

1. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein, Introduction to Algorithms, 3rd ed., MIT Press, 2009, ISBN-13: 978-0262033848.
2. Ellis Horowitz, Sartaj Sahni and Sanguthevar Rajasekaran, Fundamentals of Computer Algorithms, 1st ed., Orient Black Swan, 2008, ISBN-13: 978-8173716126.

Reference Books:

1. Robert Sedgewick, Algorithms, 4th ed., Addison-Wesley Professional, 2011, ISBN-13: 978-0321573513.
2. Steve Skiena, The Algorithm Design Manual, 2nd ed., Springer, 2008, ISBN-13: 978-1849967204.
3. Robert L. Kruse, Alexander J. Ryba, Data Structures and Program Design in C++, 1st ed., Prentice Hall, 1998, ISBN-13: 978-0137689958.
4. Mark Allen Weiss, Data Structure and Algorithm Analysis in C++, 4th ed., Pearson, 2013, ISBN-13: 978-0132847377.
5. Jeff Edmonds, How to Think About Algorithms, 1st ed., Cambridge University Press, 2014, ISBN-13: 978-1107439887.
6. Steven Halim and Felix Halim, Competitive Programming 3, 3rd ed., Lulu.com, 2013

7. Donald Ervin Knuth, The Art of Computer Programming, 3rd ed., Addison-Wesley Professional, 1997, ISBN-13: 978-020189683.
8. Donald Ervin Knuth, The Art of Computer Programming, 3rd ed., Addison-Wesley Professional, 1997, ISBN-13: 978-0321751041.